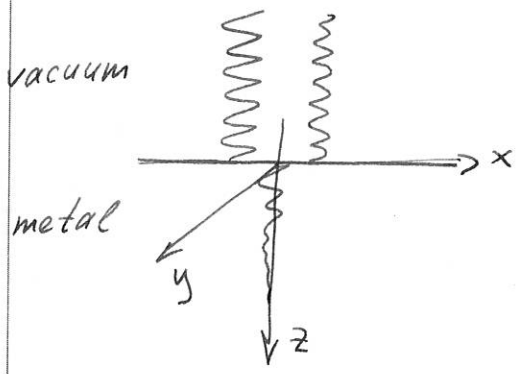
Problem 1



Incident + reflected wave

$$B_y(z<0) = B_0 e^{-i\omega t + \frac{i\omega}{c} z} + B_r e^{-i\omega t - \frac{i\omega}{c} z} \qquad (1)$$

Transmitted (evanescent) wave

$$B_y(z>0) = B_t e^{-i\omega t} e^{i\frac{\omega}{c} n z} \qquad (2)$$

where $n = n' + i n'' = (1+i)\sqrt{\frac{\sigma}{2\varepsilon_0 \omega}}$ (see Lecture notes)

Note that when σ is purely real, $n' = n''$.

To find dissipation we first need to find electric field $\vec{E} = (E_x, 0, 0)$ inside the metal

$$(\nabla \times \vec{E})_y = i\omega B_y$$

$$\frac{\partial E_x}{\partial z} = i\omega B_y$$

which togethe with Eq. (2) gives

$$E_x(z>0) = \frac{c B_t}{n} e^{-i\omega t + \frac{i\omega}{c} n z}$$

From Lecture notes $B_t = \frac{2n B_0}{1+n}$ (B_0 is real)

Taking the real part

$$E_x = 2cB_0 \, \mathrm{Re} \, \frac{e^{-i\omega t + \frac{i\omega n}{c} z}}{1+n} =$$

$$= 2cB_0 \, \mathrm{Re} \, \frac{(1+n'-in'')e^{-i\omega t + \frac{i\omega n}{c} z}}{(1+n')^2 + n''^2}$$

$$E_x = 2cB_0 \frac{(1+n')\cos(\omega t - \frac{n'\omega}{c}z) - n''\sin(\omega t - \frac{n'\omega}{c}z)}{(1+n')^2 + n''^2} e^{-\frac{\omega n''}{c}z}$$

Joule heat (per unit area) is

$$Q = \sigma_0 \int_0^\infty dz\, \overline{E_x^2(z,t)}$$

Averaging over time is performed using

$$\overline{\cos^2\omega t} = \overline{\sin^2\omega t} = \frac{1}{2}$$

$$\overline{\cos\omega t \cdot \sin\omega t} = 0$$

and yields

$$Q = 4c^2B_0^2\sigma_0 \frac{c}{2\omega n''} \frac{1}{(1+n')^2 + n''^2} \qquad (3)$$

The incident energy flux is

$$S = \frac{E_0 B_0}{\mu_0} = \frac{cB_0^2}{\mu_0} \qquad (4)$$

So far our discussion leading to Eq. (3) has been completely general. Let us now take into account that in our model $n' = n''$ and that $\sigma_0 = 2\varepsilon_0\omega n''^2$

Eq. (3) now gives

$$Q = 4c^2B_0^2 \frac{\varepsilon_0 n'' c}{(1+n'')^2 + n''^2} = \frac{cB_0^2}{\underbrace{\mu_0}_{S}} \frac{\overset{1}{\cancel{4c^2\mu_0\varepsilon_0}}\, n''}{(1+n'')^2 + n''^2}$$

or

$$\boxed{\frac{Q}{S} = \frac{4n''}{1+2n''+2n''^2}}$$

Comparing now this with R from the notes (when $n' = n''$):

$$1 - R = 1 - \frac{(n'-1)^2 + n''^2}{(n'+1)^2 + n''^2} = \frac{4n'}{(1+n')^2 + n''^2}$$

$$= \frac{4n''}{1 + 2n'' + 2n''^2} \equiv \frac{Q}{S}$$

P2

Ferrite

$$\mu_{ij} = \begin{pmatrix} \mu_\perp & -i\mu_a & 0 \\ i\mu_a & \mu_\perp & 0 \\ 0 & 0 & \mu_\parallel \end{pmatrix} \qquad \vec{B} = \mu \vec{H} \Rightarrow B_i = \mu_{ij} H_j$$

$$\vec{D} = \varepsilon \vec{E}$$

Wave equation : $\vec{\nabla} \times \vec{H} = \frac{\partial \vec{D}}{\partial t} = \varepsilon \frac{\partial \vec{E}}{\partial t}$

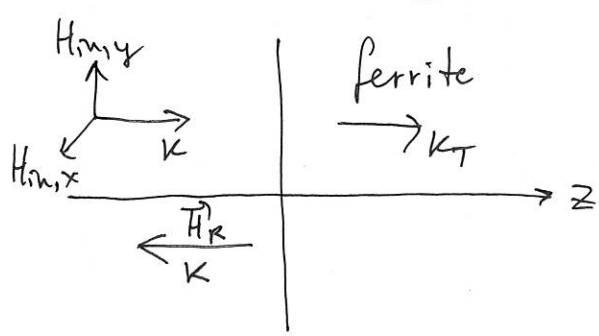
$$\underbrace{\vec{\nabla} \times (\vec{\nabla} \times \vec{H})}_{\vec{\nabla}(\vec{\nabla}\cdot\vec{H}) - \nabla^2 \vec{H}} = \varepsilon \frac{\partial}{\partial t} \underbrace{\vec{\nabla} \times \vec{E}}_{-\frac{\partial \vec{B}}{\partial t}} = -\varepsilon \frac{\partial^2}{\partial t^2} \vec{B}$$

Transverse wave : $\vec{\nabla} \cdot \vec{H} = 0$

$$\Rightarrow \frac{d^2}{dz^2} H_i = -\varepsilon \omega^2 B_i = -\varepsilon \omega^2 \mu_{ij} H_j$$

let $\vec{H}_T = (H_x, H_y, 0)\, e^{i k_T z}$ for $z > 0$.

$z < 0$ $\begin{cases} \text{Incident wave (Circularly polarized)} \quad \vec{H}_{in} = (H_{in}, i\nu H_{in}, 0)\, e^{ikz} \quad (\nu = \pm 1) \\ \text{Reflected} \quad \vec{H}_R = (H_{Rx}, H_{Ry}, 0)\, e^{-ikz} = H_R (1, i\nu, 0) e^{-ikz} \end{cases}$



$$\underbrace{k_T^2 H_j}_{k_T^2 \begin{pmatrix} H_x \\ H_y \end{pmatrix}} = \varepsilon \omega^2 \mu_{je} H_e = \varepsilon \omega^2 \begin{pmatrix} \mu_\perp & -i\mu_a \\ i\mu_a & \mu_T \end{pmatrix} \begin{pmatrix} H_x \\ H_y \end{pmatrix}$$

$$\begin{cases} (k_T^2 - \varepsilon\mu_\perp\omega^2)\, H_x = -i\mu_a \varepsilon\omega^2 H_y \\ (k_T^2 - \varepsilon\mu_\perp\omega^2)\, H_y = i\mu_a \varepsilon\omega^2 H_x \end{cases}$$

$$\Rightarrow \quad (k_T^2 - \varepsilon\mu_\perp\omega^2)^2 = (\mu_a \varepsilon\omega^2)^2$$

$$\Rightarrow \quad k_\pm^2 = k_T^2 = \varepsilon\mu_\perp\omega^2 \pm \mu_a\varepsilon\omega^2$$

For $k_+ = \sqrt{\varepsilon}\,\omega\sqrt{\mu_\perp + \mu_a}$: $\quad H_x = -i H_y$

$$\Rightarrow \quad \vec{H}_+ = H_+ (1, i, 0)\, e^{ik_+ z}$$

For $k_- = \sqrt{\varepsilon}\,\omega\sqrt{\mu_\perp - \mu_a}$: $\quad H_x = i H_y$

$$\Rightarrow \quad \vec{H}_- = H_- (1, -i, 0)\, e^{ik_- z}$$

eigenmodes are circularly polarized waves

<u>BC</u> $\quad H_{x,y}\big|_{z=-0} = H_{x,y}\big|_{z=+0} \Rightarrow (H_x \pm iH_y)\big|_{z=-0} = (H_x \pm iH_y)\big|_{z=+0}$

$\Downarrow$

circularly polarized wave basis can be used to describe continuity of $\vec{H}$.

$\Rightarrow$ if incoming wave is $\nu = +1$

$$\vec{H}_{in} = H_{in}(1, i, 0)\, e^{ikz}$$

then reflected and transmitted waves have the <u>same</u> circular polarization. Similarly for $\nu = -1$.

Thus $H_{in} + H_R = H_T$ (1)

Need another equ: continuity of $\vec{E}_{\parallel}$

$$-i\omega\varepsilon\vec{E} = \vec{\nabla}\times\vec{H} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ H_x & H_y & 0 \end{vmatrix} = \hat{x}\left(-\partial_z H_y\right) + \hat{y}\,\partial_z H_x$$

since $H_{x,y} = H_{x,y}(z)$.

$\Rightarrow$ continuity of $E_{x,y}$ gives:

For $z=+0$: $\left.\dfrac{dH_y}{dz}\right|_{z=+0} = H_\pm(\pm i)(i k_\pm)$

$$\left.\frac{dH_x}{dz}\right|_{z=+0} = H_\pm(i k_\pm)$$

So, for $z<0$ $\vec{H} = \vec{H}_{in} + \vec{H}_R =$

$$= H_{in}(1, i\nu, 0)\,e^{ikz} + H_R(1, i\nu, 0)\,e^{-ikz}$$

$$\Rightarrow \left.\frac{d}{dz}H_x\right|_{z=-0} = H_{in}(ik) + H_R(-ik)$$

$$\left.\frac{d}{dz}H_y\right|_{z=-0} = H_{in}(i\nu)(ik) + H_R(i\nu)(-ik)$$

$$\frac{(ik)(H_{in}-H_R)}{-i\omega\varepsilon_0} = \frac{H_\pm(ik_\pm)}{-i\omega\varepsilon}$$

continuity of the y-component gives the same equation, just multiplied by $(\pm i)$

$$\Rightarrow \begin{cases} \frac{K}{\varepsilon_0}\left(H_m - H_R\right) = \frac{K_\pm}{\varepsilon} H_\pm & (2) \\ H_m + H_R = H_\pm & (1) \end{cases}$$

In order not to be confused by $(\pm)$ indices, let us consider $\nu = +1$ first:

$$\begin{cases} H_m + H_R = H_+ \\ \frac{K}{\varepsilon_0}\left(H_m - H_R\right) = \frac{K_+}{\varepsilon} H_+ = \frac{K_+}{\varepsilon}\left(H_{in} + H_R\right) \end{cases}$$

$$\Rightarrow \left(\frac{K}{\varepsilon_0} - \frac{K_+}{\varepsilon}\right) H_m = \left(\frac{K}{\varepsilon_0} + \frac{K_+}{\varepsilon}\right) H_R$$

$$H_R = \frac{\frac{K}{\varepsilon_0} - \frac{K_+}{\varepsilon}}{\frac{K}{\varepsilon_0} + \frac{K_+}{\varepsilon}} H_m$$

$$\Rightarrow H_T = H_+ = H_R + H_m = \frac{2\frac{K}{\varepsilon_0}}{\frac{K}{\varepsilon_0} + \frac{K_+}{\varepsilon}} H_m .$$

Energy flux: $\vec{S} = \frac{1}{2} \vec{E} \times \vec{H}^*$

$$\Rightarrow S_m = \frac{1}{2}\left(E_{in,x} H^*_{m,y} - E_{in,y} H^*_{m,x}\right)$$

$$E_{in,x} = \frac{1}{i\omega\varepsilon_0} \frac{dH_{in,y}}{dz} = \frac{iK(iH_m)}{i\omega\varepsilon_0} = \frac{iH_m K}{\omega\varepsilon_0}$$

(because $(H_m)_y = i H_m$).

$$E_{m,y} = -\frac{1}{i\omega\varepsilon_0} \frac{dH_{m,x}}{dz} = -\frac{iKH_m}{i\omega\varepsilon_0} = -\frac{K}{\omega\varepsilon_0} H_m$$

Therefore $S_m = \frac{1}{2}\left\{ \frac{i H_m K}{\omega \varepsilon_0} (i H_m)^* - \right.$

$$\left. - \left(- \frac{K H_m}{\omega \varepsilon_0}\right) H_m^* \right\} = \frac{K}{\omega \varepsilon_0} |H_m|^2$$

Obviously $S_R = \frac{-K}{\omega \varepsilon_0} |H_R|^2$

$\Rightarrow$ reflection coefficient is

$$R_+ = \frac{|S_R|}{|S_m|} = \left|\frac{H_R}{H_m}\right|^2 = \left(\frac{\frac{K}{\varepsilon_0} - \frac{K_+}{\varepsilon}}{\frac{K}{\varepsilon_0} + \frac{K_+}{\varepsilon}}\right)^2 .$$

$$K = \frac{\omega}{c}, \quad K_+ = \omega \sqrt{\varepsilon} \sqrt{\mu_\perp + \mu_a}$$

$$\frac{1}{c} = \sqrt{\varepsilon_0 \mu_0} \Rightarrow \frac{K}{\varepsilon_0} = \omega \frac{\sqrt{\varepsilon_0 \mu_0}}{\varepsilon_0} = \sqrt{\frac{\mu_0}{\varepsilon_0}}\, \omega$$

$$\frac{K_+}{\varepsilon} = \omega \sqrt{\frac{\mu_\perp + \mu_a}{\varepsilon}}$$

$$\Rightarrow R_+ = \left(\frac{\sqrt{\frac{\mu_0}{\varepsilon_0}} - \sqrt{\frac{\mu_\perp + \mu_a}{\varepsilon}}}{\sqrt{\frac{\mu_0}{\varepsilon_0}} + \sqrt{\frac{\mu_\perp + \mu_a}{\varepsilon}}}\right)^2$$

is expressed via impendance $Z = \sqrt{\frac{\mu}{\varepsilon}}$

Obviously, $R_- = \left(\frac{\sqrt{\frac{\mu_0}{\varepsilon_0}} - \sqrt{\frac{\mu_\perp - \mu_a}{\varepsilon}}}{\sqrt{\frac{\mu_0}{\varepsilon_0}} + \sqrt{\frac{\mu_\perp - \mu_a}{\varepsilon}}}\right)^2 .$

In terms of $\tilde{\varepsilon} = \frac{\varepsilon}{\varepsilon_0}$, $\tilde{\mu}_{\perp,a} = \frac{\mu_{\perp,a}}{\mu_0}$ can write

$$R_\pm = \left(\frac{1 - \sqrt{\frac{\tilde{\mu}_\perp \pm \tilde{\mu}_a}{\tilde{\varepsilon}}}}{1 + \frac{\tilde{\mu}_\perp \pm \tilde{\mu}_a}{\tilde{\varepsilon}}}\right)^2$$

Now, $R_\pm = 1$ when $\sqrt{\dfrac{\tilde{\mu}_\perp \pm \tilde{\mu}_a}{\tilde{\varepsilon}}} = \infty$ or 0.

$$\tilde{\mu}_\perp \pm \tilde{\mu}_a = 1 + \frac{\omega_M \omega_0}{\omega_0^2 - \omega^2} \pm \frac{\omega_M \omega}{\omega_0^2 - \omega^2} = 1 + \frac{\omega_M(\omega_0 \pm \omega)}{\omega_0^2 - \omega^2}$$

$$= 1 + \frac{\omega_M}{\omega_0 \mp \omega} = \frac{\omega_0 \mp \omega + \omega_M}{\omega_0 \mp \omega}$$

Therefore $\tilde{\mu}_\perp + \tilde{\mu}_a = \infty$ when $\omega = \omega_0$ (resonance)

since $\tilde{\mu}_\perp + \tilde{\mu}_a = 1 + \dfrac{\omega_M}{\omega_0 - \omega} = \dfrac{\omega_0 + \omega_M - \omega}{\omega_0 - \omega} \to \infty$ $\omega \to \omega_0$.

Observe that $\tilde{\mu}_\perp + \tilde{\mu}_a = 0$ for $\omega = \omega_0 + \omega_M$.

$$\Rightarrow R_+(\omega = \omega_0) = 1, \quad R_+(\omega = \omega_0 + \omega_M) = 1.$$

But $\tilde{\mu}_\perp - \tilde{\mu}_a = 1 + \dfrac{\omega_M}{\omega_0 + \omega} = \dfrac{\omega_0 + \omega_M + \omega}{\omega_0 + \omega}$

is always finite (and does not diverge) for physical $\omega > 0$.

Therefore reflection for $V = -1$ is never complete,

$R_-(\omega) < 1$ for all ω.